

A New Proof of an Arithmetic Theorem

English translation of Gauss, Werke, Band II, Articles 5–7

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Translation prepared from the source-locked TeX retype in this packet

5.

THEOREM. *Let x be a positive non-integral quantity, among whose multiples $x, 2x, 3x, \dots$ up to nx no integer occurs; put $[nx] = h$. From this it is readily concluded that among the multiples of the reciprocal quantity $\frac{1}{x}, \frac{2}{x}, \frac{3}{x}, \dots$ up to $\frac{h}{x}$ no integer is found either. Then I say that*

$$\left. \begin{aligned} & [x] + [2x] + [3x] + \dots + [nx] \\ & + \left[\frac{1}{x}\right] + \left[\frac{2}{x}\right] + \left[\frac{3}{x}\right] + \dots + \left[\frac{h}{x}\right] \end{aligned} \right\} = nh.$$

Proof. The members of the series $[x] + [2x] + [3x] + \dots + [nx]$, which we shall put equal to Ω , are manifestly all equal to 0 from the first up to and including $\left[\frac{1^{\text{st}}}{x}\right]$; all the following members up to $\left[\frac{2^{\text{nd}}}{x}\right]$ are equal to 1; all the following members up to $\left[\frac{3^{\text{rd}}}{x}\right]$ are equal to 2; and so on. Hence

$$\Omega = \left. \begin{aligned} & 0 \times \left[\frac{1}{x}\right] \\ & + 1 \times \left\{ \left[\frac{2}{x}\right] - \left[\frac{1}{x}\right] \right\} \\ & + 2 \times \left\{ \left[\frac{3}{x}\right] - \left[\frac{2}{x}\right] \right\} \\ & + 3 \times \left\{ \left[\frac{4}{x}\right] - \left[\frac{3}{x}\right] \right\} \\ & \text{etc.} \\ & + (h-1) \left\{ \left[\frac{h}{x}\right] - \left[\frac{h-1}{x}\right] \right\} \\ & + h \left\{ n - \left[\frac{h}{x}\right] \right\} \end{aligned} \right\} = hn - \left[\frac{1}{x}\right] - \left[\frac{2}{x}\right] - \left[\frac{3}{x}\right] - \dots - \left[\frac{h}{x}\right].$$

Q.E.D.

6.

THEOREM. *If k, p denote any positive odd numbers prime to one another, then*

$$\left. \begin{aligned} & \left[\frac{k}{p}\right] + \left[\frac{2k}{p}\right] + \left[\frac{3k}{p}\right] + \dots + \left[\frac{\frac{1}{2}(p-1)k}{p}\right] \\ & + \left[\frac{p}{k}\right] + \left[\frac{2p}{k}\right] + \left[\frac{3p}{k}\right] + \dots + \left[\frac{\frac{1}{2}(k-1)p}{k}\right] \end{aligned} \right\} = \frac{1}{4}(k-1)(p-1).$$

Proof. Suppose, as is permissible, that $k < p$. Then $\frac{\frac{1}{2}(p-1)k}{p}$ will be less than $\frac{1}{2}k$, but greater than $\frac{1}{2}(k-1)$, and therefore $\left[\frac{\frac{1}{2}(p-1)k}{p}\right] = \frac{1}{2}(k-1)$. Hence it is plain that the present theorem

follows at once from the preceding one, by putting there $\frac{k}{p} = x$, $\frac{1}{2}(p-1) = n$, and therefore $\frac{1}{2}(k-1) = h$.

Moreover, in a similar way it can be proved that, if k is an *even* number prime to p , then

$$\left. \begin{aligned} & \left[\frac{k}{p} \right] + \left[\frac{2k}{p} \right] + \left[\frac{3k}{p} \right] + \cdots + \left[\frac{\frac{1}{2}(p-1)k}{p} \right] \\ & + \left[\frac{p}{k} \right] + \left[\frac{2p}{k} \right] + \left[\frac{3p}{k} \right] + \cdots + \left[\frac{\frac{1}{2}kp}{k} \right] \end{aligned} \right\} = \frac{1}{4}k(p-1).$$

But we shall not pause over this proposition, since it is unnecessary for our purpose.

7.

Now, by combining the preceding theorem with Proposition VIII of Article 4, the fundamental theorem follows at once. Namely, if k and p denote any unequal positive prime numbers, and if one puts

$$(k, p) + \left[\frac{k}{p} \right] + \left[\frac{2k}{p} \right] + \left[\frac{3k}{p} \right] + \cdots + \left[\frac{\frac{1}{2}(p-1)k}{p} \right] = L$$

$$(p, k) + \left[\frac{p}{k} \right] + \left[\frac{2p}{k} \right] + \left[\frac{3p}{k} \right] + \cdots + \left[\frac{\frac{1}{2}(k-1)p}{k} \right] = M,$$

then by Proposition VIII of Article 4 it is clear that L and M always become even numbers. But by the theorem of Article 6,

$$L + M = (k, p) + (p, k) + \frac{1}{4}(k-1)(p-1).$$

Therefore, whenever $\frac{1}{4}(k-1)(p-1)$ turns out even, which happens if both k, p , or at least one of them, is of the form $4n+1$, the numbers (k, p) and (p, k) must necessarily be either both even or both odd. Whenever, on the other hand, $\frac{1}{4}(k-1)(p-1)$ is odd, which occurs if both k, p are of the form $4n+3$, one of the two numbers $(k, p), (p, k)$ must necessarily be even and the other odd. Thus, in the former case the relation of k to p and the relation of p to k - insofar as one is or is not a residue of the other - will be identical; in the latter case they will be opposite. Q.E.D.

Translation note: bracket notation is preserved as in the source TeX. The symbols (k, p) and (p, k) are retained unchanged because the argument refers back to Article 4, Proposition VIII, where their parity interpretation is fixed.